

## TERMS OF REFERENCE

### AI-Driven Tender Query Automation and Pre-Bid Reply Generation System

#### 4. Introduction

##### 4.1 Problem Statement

- a. Manual RFP analysis is time-consuming and can result in conflicting inferences.
- b. Vendors struggle to identify missing/ambiguous clauses in RFP document spanning hundreds of Pages.
- c. Vendors put up pre-bid queries without fully going through the voluminous RFP. This leads to Unnecessary repetition of queries and similarity of replies thereof.
- d. Department lack automated conformance checks of RFP documents before release and analyse various provisions and cross references in case of any ambiguity persists.

##### 4.2 Purpose

Procurement processes typically involve publishing Request for Proposals (RFPs) or Standard Bidding Documents (SBDs), followed by pre-bid meetings and clarifications sought by bidders. Responding to these queries manually is time-consuming, prone to inconsistencies, and lacks historical context.

To enhance efficiency, accuracy, and consistency in responding to tender queries, an AI-driven system is proposed to automate pre-bid reply generation based on:

- RFPs / Tender Documents
- Standard Bidding Documents (SBD) templates
- Historical records of similar query replies

The solution shall support both cloud-based and on-premise deployments, offering flexibility based on organizational infrastructure. It should enhance the tender analysis process by providing instant insights, efficient query handling, and both speech-to-text and traditional input methods.

##### 4.3 Objective

To design, develop, and deploy an AI-driven solution capable of: -

- a. Automatically parsing and understanding tender documents and bidder queries
- b. Suggesting or generating appropriate draft responses to queries
- c. Leveraging past query-response data for consistency
- d. Minimizing repetitive manual effort through AI assistance, while enabling human oversight in response approval
- e. Enhancing the bidder and admin experience

##### 4.4 Scope of Work

- a. A web-based application with an LLM backend for AI-Power Tender Query Automation system.
- b. Vendors to register through NHAI Web site to access the envisaged system under the RFP.
- c. Login and raise queries against a specific RFP already hosted by website of NHAI.
- d. Provide AI-driven insights (missing info, anomalies and clarifications).

- e. Maintain a query history for vendors.
- f. Offer an admin panel for analytics, conformance checks, and corrigendum generation and query history to alert NHAI on any ambiguity raised by bidders.

#### **4.5 Document Ingestion & Knowledge Base Creation**

- a. Development of a module to ingest and parse RFPs, Standard Bidding Document (SBD), Policy Circular, Corrigenda, and tender annexures in multiple formats (PDF, DOCX, TXT).
- b. Building a structured, searchable knowledge base from historical query responses.

#### **4.6 AI-Powered Query Classification & matching**

- a. Implementation of Natural Language Processing (NLP) models to classify queries by type (eligibility, technical specs, and commercial, contractual)
- b. Retrieval of similar past queries and corresponding responses.

#### **4.7 Draft Reply Generation using AI/ML techniques**

- a. Develop AI / ML models to auto-generate suggested replies.
- b. Highlight the reference section from the current tender/RFP or past records.
- c. Speech-to-text enabled query input.

#### **4.8 User Interface for Vendors.**

- a. Registration, login capability.
- b. Upload specific RFPs for analysis.
- c. Query submission (keyboard / speech-to-text)
- d. Real-time chatbot support using the underlying LLM for vendor assistance.
- e. Access historical queries and receive categorized insights (missing info, anomalies).

#### **4.9 User interface for NHAI -Admin**

- a. Role-based user access control.
- b. Dashboard for tracking query status, turnaround time, and frequently asked queries.
- c. Auto-conformance check of RFPs with standard templates/documents.
- d. Analytics on query trends, bidder concerns, and recurring clarifications.
- e. Generate draft replies to the queries raised by the bidders.
- f. Provide audit trail and version control.
- g. AI-assisted corrigendum generation.
- h. Generate compliance reports for internal audit and regulatory checks.

#### **4.10 Integration & Compatibility**

- a. REST API support for LLM (e.g., GPT-4 or LLaMA 3).
- b. Integration with meeting platforms (e.g., WebEx).
- c. Email-based approval workflows.
- d. Hyperlink support within documents.

#### **4.11 Other Deliverables**

- a. Integrated knowledge base of tender documents and past responses.

- b. User manuals and operational documentation.
- c. Training sessions for end-users (NHA officers).
- d. Technical documentation and API integrations.

#### 4.12 Technology Stack (Indicative)

- a. AI/ML: NLP models (GPT-based fine-tuned models)
- b. Backend: Python (FastAPI or Django) / Java / Node.js
- c. Frontend: React / Angular / Vue.js
- d. Database: PostgreSQL / MongoDB / Elasticsearch
- e. Deployment: On-premise / Cloud

Preferred stack is Python (FastAPI), PostgreSQL, and React. Vendors may propose alternatives with justification and performance benchmarks.

### 5. Functional Requirements

| Category               | Requirement                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Authentication         | Multi-factor login (SMS/email), Password Recovery, Role-based user access (Vendor/Admin)                              |
| Query Handling         | NLP-based classification (eligibility, specs, commercial, legal)                                                      |
| Reply Generation       | Auto-generated draft replies using LLM with source highlights                                                         |
| Interaction            | Upload RFP for analysis, Interactive Q&A chatbot for vendors                                                          |
| Analytics              | Dashboard for trends, bottlenecks, and compliance tracking for admin                                                  |
| Conformance Check      | Clause-by-clause compliance with standard SBD templates                                                               |
| Corrigendum Generation | AI-Assisted Corrigendum Generation                                                                                    |
| Workflow               | Approval versioning and audit trail                                                                                   |
| Integration            | WebEx, email workflows, REST APIs                                                                                     |
| Document Handling      | Upload any format (PDF, DOCX, PPTX), render to PDF                                                                    |
| AI and LLM Integration | LLM fine-tuned for legal & procurement language. Automatically detect missing clauses and policy deviations. (if any) |
| Chat bot               | Real time querying on RFP document.                                                                                   |

### 6. Performance Requirements

- a. <15 seconds AI response at 25% load
- b. <50 seconds AI response at 100% load
- c. <2 seconds response for cached queries
- d. Support for 100 concurrent users
- e. System uptime of 99.5% per month (excluding scheduled maintenance).
- f. System must support scaling to 500 concurrent users with minimal performance degradation.

### 7. Security & Compliance

- a. Compliance with India's DPDP guidelines.
- b. AES-256 encryption at rest; TLS 1.3 in transit.
- c. Zero-trust security model.
- d. The LLM shall operate in stateless mode. No tender data will be stored or retained by the LLM

beyond session scope. All historical query and response data will be securely stored in a structured knowledge base managed by the application backend for future reference and learning purposes, in compliance with data protection standards.

#### 8. Implementation Timelines (Total 8 weeks)

| Phase                                           | Duration |
|-------------------------------------------------|----------|
| Requirement Gathering & Concept Design Document | 2 week   |
| Development & Integration                       | 3 weeks  |
| UAT & Training                                  | 2 week   |
| Go-Live                                         | 1 week   |

Post Go-Live the service provider shall maintain the system for a period of 3 years. Resources to be deployed by the service provider at NHAHQ New Delhi with minimum qualification & experience is as under:

| S.No. | Designation                     | Nos | Educational Qualification                                                                              | Professional Experience                                                                                                                                                                                                               | Deployment                                                    |
|-------|---------------------------------|-----|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 1     | Project Manager/<br>Team Leader | 1   | Graduate/Postgraduate in Computer Science/IT/Engineering from a recognized University/ Institute       | Minimum 5 years' experience in managing IT/e- Governance/ Enterprise Software projects, OR<br>Experience in leading implementation of similar Projects.<br><i>Preference will be given to project of similar nature / complexity.</i> | Onsite till go-live, thereafter ten (10) man days per months. |
| 2     | AI/LLM Specialist               | 2   | Graduate/Postgraduate in Computer Science, Data Science, AI/ML from a recognized University/ Institute | Minimum 3 years' experience in developing and integrating LLMs/NLP systems for document processing. Hands-on experience <b>with</b> local LLM deployment, fine-tuning, and inference pipelines.                                       | Onsite                                                        |
| 3     | Backend Developer               | 2   | Graduate in Computer Science/IT Engineering or related discipline                                      | At least 3 years' experience in backend development using Node.js/Python.- Experience in building secure APIs, file upload/download mechanisms, and integrating ML models.                                                            | Onsite                                                        |
| 4     | Business Analyst                | 1   | Graduate in Business Administration/IT/Engineering                                                     | Minimum 3 years of experience in requirement gathering and documentation for IT systems. Understanding of public procurement/tender workflows is preferable.                                                                          | Onsite                                                        |

## 9. Confidentiality & Data Security

- a. The system should comply with applicable data protection guidelines and ensure secure handling of sensitive procurement documents and bidder information.
- b. All data must reside within India. No cloud data shall be hosted outside national boundaries unless explicitly approved by NHAI.
- c. All source code, training data, AI models (including fine-tuned LLMs), APIs, and documentation developed as part of this engagement will be the property of NHAI. The vendor shall not reuse, resell, or replicate the system for any third party without prior written approval.

## 10. Future Scalability and Upgrades

The system must be designed with modularity and extensibility to accommodate future upgrades, including but not limited to:

- a. Integration of newer, more capable LLMs (e.g., future GPT/LLaMA versions)
- b. Automated confidence scoring and tiered response approvals, where high-confidence, low-risk AI-generated replies may be auto-approved with audit logging and admin override
- c. Expansion of concurrent user capacity beyond 500 users
- d. Multilingual query support
- e. Real-time integration with e-procurement systems
- f. Mobile app extension for query submission and status tracking
- g. Enhanced analytics and reporting modules (custom dashboards, exportable formats)
- h. Conformance with evolving data protection and procurement regulations

## 11. Definition & Acronyms

| Term                          | Definition                                                                                                                                                                                                                                                                                                      |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RFP</b>                    | Request for Proposal                                                                                                                                                                                                                                                                                            |
| <b>LLM</b>                    | Large Language Model (e.g., GPT-4, Claude, etc.)                                                                                                                                                                                                                                                                |
| <b>Corrigendum</b>            | An amendment to a tender document                                                                                                                                                                                                                                                                               |
| <b>Conformity Level Check</b> | For any new RFP, the level of conformity with the standard official documents and policies which are brought out from time to time. This check will provide clause by clause conformance validation, overall percentage and sections that are anomalous, conflicting with the published documents and policies. |
| <b>NLP</b>                    | Natural Language Processing                                                                                                                                                                                                                                                                                     |
| <b>SBD</b>                    | Standard Bidding Document                                                                                                                                                                                                                                                                                       |
| <b>MFA</b>                    | Multi-Factor Authentication                                                                                                                                                                                                                                                                                     |
| <b>TAT</b>                    | Turnaround Time                                                                                                                                                                                                                                                                                                 |

## 12. System Architecture

### a. High-Level Design [Indicative]

- i. **Frontend:** React.js
- ii. **Backend:** Python (Django/Fast API)
- iii. **LLM:** OpenAI GPT-4 / Llama 3 (self-hosted)
- iv. **Database:** PostgreSQL

**b. Technology Stack**

| Component | Technology             |
|-----------|------------------------|
| Frontend  | React, Bootstrap       |
| Backend   | FastAPI, Django        |
| AI Model  | GPT-4 / Fine-tuned LLM |
| Database  | PostgreSQL             |

**13. Data Requirements**

- a. Data Inputs**
  - i. RFP documents (PDF/DOCX), Standard Bidding Document (SBD), Policy Circular, Corrigenda, and tender annexures in multiple formats (PDF, DOCX, TXT)
  - ii. Vendor queries (text).
- b. Data Outputs**
  - i. AI analysis report (JSON/CSV/PDF).
  - ii. Corrigendum drafts (CSV).

- 14. Assumptions & Constraints**
- i. Assumption: Queries will be made within the context of the RFP. The LLM will also limit the context within this purview
  - ii. Constraint: LLM will not store sensitive tender data beyond session.

**15. Risks & Mitigation Strategies**

| Risk                 | Mitigation                   |
|----------------------|------------------------------|
| AI misinterpretation | Human-in-the-loop validation |
| Data breach          | Zero-trust security model    |